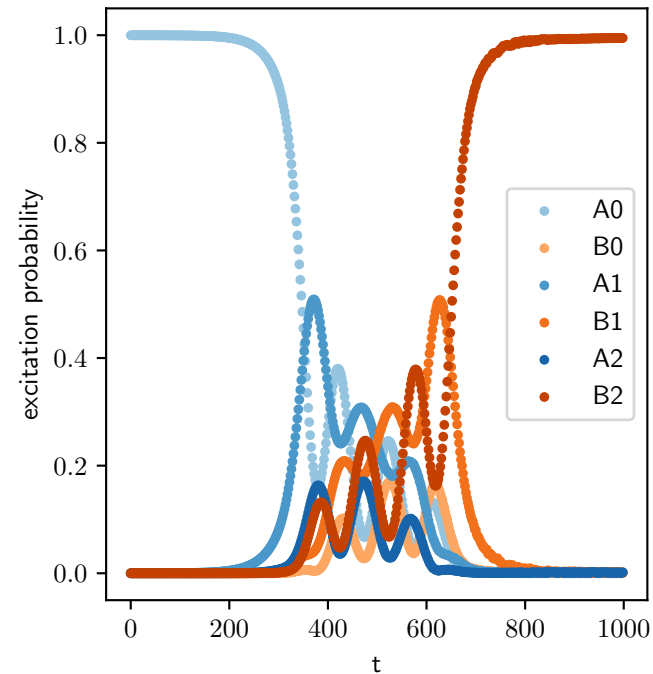


spin transport



phonon transport

